

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across different product categories. The goal is to identify spending patterns, customer segments, product preferences, discount usage, subscription behavior, and age-based revenue patterns to support business decisions.

2. Dataset Summary

Rows: 3,900

Columns: 18

Key Features:

- Customer demographics (Age, Gender, Location)
- Purchase details (Item Purchased, Category, Purchase Amount, Size, Color, Season)
- Shopping behavior (Discount Applied, Previous Purchases, Payment Method, Frequency of Purchases)
- Customer feedback and services (Review Rating, Subscription Status, Shipping Type)

Missing Data: 37 values in the Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` and `df.describe(include='all')` to inspect structure and summary statistics.
- **Missing Data Handling:** Filled missing Review Rating values using the median rating of each product category.

3. Exploratory Data Analysis using Python (continued)

- **Column Standardization:** Converted column names to lowercase and replaced spaces with underscores.
- **Feature Engineering:** Created an age_group column by categorizing customer ages.
- **Purchase Frequency:** Created purchase_frequency_days by converting purchase-frequency categories into approximate days.
- **Data Consistency Check:** Verified that discount_applied and promo_code_used contained the same information and removed the redundant column.

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
df = df.drop('promo_code_used', axis=1)
```

Python data loading and initial exploration

```
import pandas as pd
```

```
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Visa
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	MasterCard
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Visa
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	MasterCard
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Visa

```
df.info()
```

Descriptive statistics and data structure

3. Exploratory Data Analysis using Python (continued)

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

```
df.isnull().sum()
```

Missing-value treatment and column standardization

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD) 0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status 0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
```

Age-group feature engineering

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

```
#create a column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged

Purchase-frequency conversion

4. Data Analysis using SQL

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male and female customers.

Gender	Revenue
Female	75,191
Male	157,890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

Result: 839 customers identified.

3. **Top 5 Products by Average Review Rating** – Identified products with the highest average customer ratings.

Rank	Product	Average Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Handbag	3.78

4. **Standard vs Express Shipping** – Compared average purchase amount between Standard and Express shipping.

Shipping Type	Average Purchase
Express	\$60.48
Standard	\$58.46

4. Data Analysis using SQL (continued)

5. **Subscribers vs Non-Subscribers** – Compared customer count, average spending, and total revenue.

Subscription	Customers	Avg. Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

6. **Products with Highest Discount Usage** – Identified products with the highest percentage of discounted purchases.

Rank	Product	Discount Rate
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

7. **Customer Segmentation** – Segmented customers into New, Returning and Loyal based on previous purchases.

Segment	No. of Customers
New	83
Returning	701
Loyal	3,116

8. **Top Products Within Each Category** – Identified the three most purchased products within each category.

- **Accessories:** Jewelry (171), Belt (161), Sunglasses (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)
- **Outerwear:** Jacket (163), Coat (161)

4. Data Analysis using SQL (continued)

9. **Repeat Buyers and Subscription** – Analyzed subscription status among customers with more than five previous purchases.

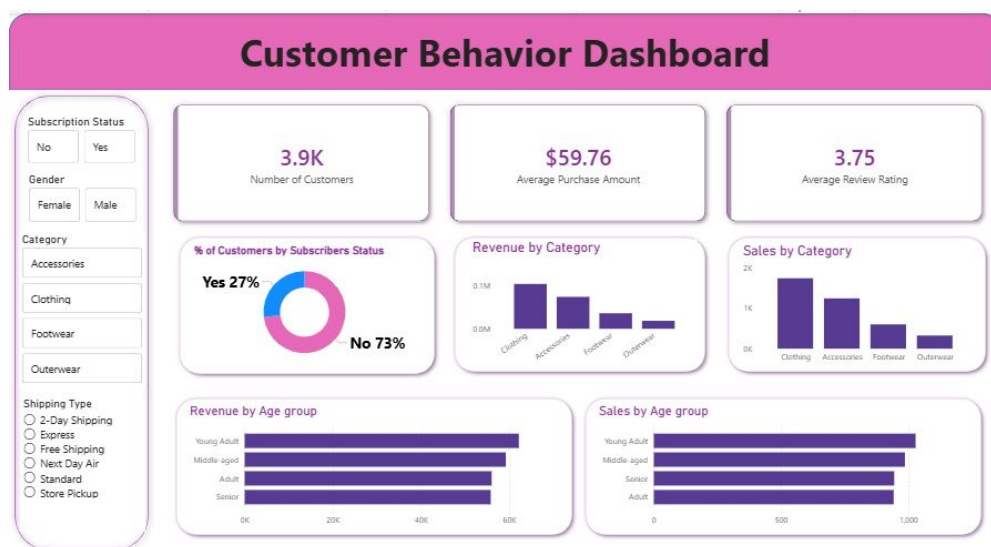
Subscription	Repeat Buyers
No	2,518
Yes	958

10. **Revenue by Age Group** – Compared total revenue generated by different customer age groups.

Age Group	Total Revenue
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

5. Customer Behavior Dashboard

Created an interactive dashboard showing customer count, average purchase amount, average review rating, subscription status, revenue and sales by category, and revenue and sales by age group. Filters were provided for subscription status, gender, category, and shipping type.



Customer Behavior Dashboard

6. Business Recommendations

- **Customer Segmentation** – Develop targeted strategies for New, Returning, and Loyal customers.
- **Product Strategy** – Focus marketing efforts on high-performing products and categories.
- **Discount Strategy** – Review discount usage to balance customer attraction with revenue.
- **Subscription Promotion** – Encourage repeat customers to join subscription programs.
- **Targeted Marketing** – Use age-group and customer behavior insights for personalized campaigns.

The analysis combines Python-based data preparation, SQL-based business analysis, and interactive dashboard visualization to provide a concise view of customer shopping behavior.